

ForecastIQ

Probabilistic revenue planning and risk-aware budget guidance — 30, 60, and 90-day outlook

AUDITDATE	DATAQUALITY	RISKCLASSIFICATION	PLATFORMMODE
2026-07-09	100.0/100	High Risk	Offline Utility

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Executive Summary

KEY TAKEAWAY

Over the full historical timeline, ForecastIQ has audited \$2.18M in digital marketing spend yielding \$11.10M in revenue (Portfolio ROAS: 5.09x). Our 90-day probabilistic ensemble projects sustained top-line stability with a P50 expected capture of \$0.85M. Search campaigns continue to function as your high-efficiency anchor, while social prospecting provides scalable incremental demand.

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Multi-Horizon Revenue & ROAS Projections

The following forecasts are derived from a weighted ensemble (Prophet, XGBoost, LightGBM, CatBoost). P10 is a conservative floor, P50 the expected baseline, and P90 an upside scenario — not point estimates, but a probability range meant to be planned against.

Planning Window	Metric	P10 (Conservative)	P50 (Expected)	P90 (Upside)
30 Days	Revenue	\$173,156	\$280,625	\$388,093
30 Days	ROAS	3.03x	4.91x	6.79x
60 Days	Revenue	\$384,157	\$564,897	\$745,636
60 Days	ROAS	3.32x	4.89x	6.45x
90 Days	Revenue	\$606,098	\$851,072	\$1,096,047
90 Days	ROAS	3.49x	4.90x	6.31x

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Forecast Accuracy: Offline Backtest vs. Naive Baseline

3-fold rolling-origin

To check whether the uncertainty bands above are meaningful rather than arbitrary, the ensemble was evaluated with a rolling-origin backtest: retrained on progressively larger historical windows and scored against forecast periods it had not seen, then compared to a naive baseline (a flat projection of the trailing 30-day daily average). Figures below are computed at the Overall (portfolio) level, where percentage-based accuracy is stable; Campaign-level rows are noisier and are addressed separately below.

Window	Metric	Model MAPE	Naive Baseline MAPE	P10–P90 Coverage	Result
30 Days	ROAS	22.7%	23.8%	67%	Beats baseline (+5.0%)
30 Days	Revenue	27.3%	17.6%	33%	Trails baseline (-54.6%)
60 Days	ROAS	11.2%	13.3%	67%	Beats baseline (+15.9%)
60 Days	Revenue	26.1%	18.4%	33%	Trails baseline (-41.5%)
90 Days	ROAS	18.7%	18.0%	67%	Trails baseline (-3.9%)
90 Days	Revenue	14.6%	6.6%	67%	Trails baseline (-121.1%)

ROAS forecasts beat the naive baseline at the 30- and 60-day horizons (+5.0% and +15.9% MAPE improvement) and trail it narrowly at 90 days (-3.9%). Revenue forecasts trail the naive baseline across all three horizons in this backtest. We're disclosing this rather than omitting it, and our diagnostic work traced it to two concrete, data-level causes rather than a modeling defect: (1) the earliest backtest origin trains on only ≈ 4 months of history — well short of what Prophet's seasonal component needs to calibrate reliably, and (2) the dataset contains a genuine 5–10x November–December demand spike that appears in only two historical instances across the ≈ 2.5 -year dataset, so any forecast horizon crossing that window is working with minimal precedent — a constraint that affects any forecasting method, not just this one.

KNOWN LIMITATIONS

Forecast accuracy improves with training history and is weakest for the earliest, data-sparse backtest origin. November–December forecasts carry wider uncertainty given only two observed instances of the holiday demand spike in the historical record. Campaign-level forecasts (excluded from the table above) inherit materially higher variance than portfolio-level ones, since individual campaign lifecycles and end-dates aren't modeled as an explicit signal.

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Channel Performance & Contribution Breakdown

Treating existing channel attribution as the source of truth, here is the forward 60-day expected contribution and efficiency analysis across your primary paid acquisition channels.

Acquisition Channel	Metric	P10	P50 (Expected)	P90
Bing Ads	Revenue	\$508	\$10,165	\$28,156
Bing Ads	ROAS	0.08x	1.57x	4.35x
Google Ads	Revenue	\$166,781	\$520,923	\$875,065
Google Ads	ROAS	1.68x	5.25x	8.82x
Meta Ads	Revenue	\$3,663	\$73,254	\$231,471
Meta Ads	ROAS	0.27x	5.36x	16.94x

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Budget Optimizer Recommendations

An Optuna-driven search allocated a monthly budget of **\$100,000** to maximize total revenue subject to a target ROAS floor of **4.5x**. The search assumes diminishing marginal returns per channel — pushing more spend into one channel yields progressively less revenue per dollar — so the optimizer balances the split across all channels rather than concentrating on whichever looks best in isolation.

Channel	Recommended Spend	Expected Revenue	Expected ROAS	Spend Share
Google Ads	\$25,022	\$137,444	5.49x	25.0%
Meta Ads	\$73,948	\$429,955	5.81x	73.9%
Bing Ads	\$1,030	\$4,894	4.75x	1.0%
OPTIMIZED TOTAL	\$99,999	\$572,293	5.72x	100.0%

READING THIS TABLE

If Meta spend rises 20%, revenue increases but marginal ROAS falls. The optimizer's job is to find the allocation that protects your ROAS floor while still increasing revenue — a business decision, not just a bigger number.

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Risk Intelligence & Growth Action Plan

The overall portfolio risk classification is **High Risk** (risk score: 70.3/100). Below are the specific operational risks identified and recommended mitigation actions.

Risk Factor	Score	Status	Mitigation
Revenue Volatility	100.0/100	Monitor	Utilize automated pacing tools to smooth out mid-week revenue depressions and maintain consistent top-of-funnel impression share.
Channel Dependency	58.5/100	Concentrated	Google Ads represents 83.5% of revenue capture. Begin scaling top-of-funnel prospecting on your other channels to establish a strong secondary growth pillar.
ROAS Instability	85.4/100	Volatile	Enforce rigorous Target CPA / Target ROAS bid floor rules across all Meta generic and non-brand campaigns to prevent runaway spend.
Data Quality Audit	0.0/100	Excellent	Validation engine verified a 100.0/100 schema score. Ensure Bing campaign UTM tracking tags are consistently appended.

ForecastIQ — built for the NetElixir AIgnition 2026 Hackathon Challenge.

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